Artificial Intelligence & Machine Learning – Task 2

Feature Engineering, Model Optimization & Performance Comparison

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Dataset: California Housing Dataset

Objective

The objective of this project was to build and compare multiple machine learning regression models for predicting California housing prices. The project focused on feature preprocessing, model optimization, and performance evaluation using standard regression metrics.

Methodology

1. Dataset Loading
The California Housing dataset was imported from the Scikit-learn library. It contains several housing-related features such as median income, house age, average number of rooms, population, latitude, and longitude. The target variable represents the median house value.

2. Data Exploration
The dataset was explored using head(), info(), describe(), and isnull().sum(). The dataset contained 20,640 records, 8 input features, and no missing values.

3. Feature Engineering
Features (X) and target (y) were separated. StandardScaler was applied to normalize feature values.

4. Data Splitting
The dataset was split into 80% training and 20% testing using train_test_split() with a fixed random_state.

5. Model Training
Linear Regression, Ridge Regression, and Decision Tree Regressor were trained and tested.

6. Model Evaluation
Models were evaluated using RMSE and R^2 Score.

Results

Model	RMSE	R^2 Score
Linear Regression	0.745581	0.575788
Ridge Regression	0.745554	0.575819
Decision Tree Regressor	0.724234	0.599732

The Decision Tree Regressor achieved the best overall performance with the lowest RMSE and the highest R^2 Score, indicating more accurate predictions compared to the linear models.

Observations

• The dataset contained no missing values.
• Feature scaling standardized all input variables before training.
• Linear Regression and Ridge Regression produced nearly identical results.
• Decision Tree Regressor achieved the lowest prediction error.
• The Actual vs Predicted plot indicated that the Decision Tree model captured the relationship between features and house prices more effectively.

Conclusion

This project demonstrated the complete machine learning workflow, including data exploration, preprocessing, feature scaling, model training, evaluation, and comparison. Among the three evaluated models, the Decision Tree Regressor produced the best predictive performance with an RMSE of 0.724234 and an R^2 Score of 0.599732.

The project highlights the importance of comparing multiple algorithms rather than relying on a single model. Proper preprocessing and evaluation are essential steps in developing reliable machine learning solutions for regression problems.